

Is This Deal Worth It? An Analytics-Driven Evaluation of Energy Sector M&A

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1. Introduction

This report examines whether major mergers and acquisitions (M&A) in the U.S. energy industry create or destroy value for acquiring companies. The project develops a logistic regression classification model that uses financial information available before an acquisition to predict potential deal outcomes.

The analysis focuses on four major Houston-area energy acquisitions: ExxonMobil's acquisition of Pioneer Natural Resources, ConocoPhillips' acquisition of Marathon Oil, Chevron's acquisition of PDC Energy, and Diamondback Energy's acquisition of Endeavor Energy Resources. These four transactions were excluded from model training and were used as unseen examples to demonstrate how the prediction framework could be applied to real-world acquisitions.

The classification model was trained using a historical dataset of 56 energy-sector M&A deals completed between 2005 and 2024. The goal of this project is not to determine whether these individual acquisitions were successful or unsuccessful, but to examine whether information available before a deal can provide useful indicators of future outcomes.

2. Business Problem Definition

Large acquisitions in the energy industry often involve investments worth billions of dollars, making them important strategic decisions for companies. However, it is difficult to know before a deal closes whether it will improve or reduce shareholder value. Most evaluations are performed after the acquisition has been completed, when companies have already committed significant financial resources. Developing a predictive model based on financial information available before the acquisition could provide decision makers with an additional source of evidence when evaluating potential transactions.

3. Problem Statement

This study examines whether four pre-acquisition financial variables, including return on assets, debt-to-equity ratio, earnings per share, and deal value, can be used to predict whether an acquisition will create or destroy value for the acquiring company. A logistic regression model was developed using historical M&A transactions and then applied to four Houston area acquisitions that were excluded from the training dataset. The goal was to determine whether these financial variables provide enough information to accurately classify acquisition outcomes.

4. Project Objectives

1. The project was designed around the following objectives:

2. Collect and prepare financial and operational data for selected U.S. energy-sector acquisitions using SEC EDGAR filings and other publicly available sources.
3. Conduct exploratory and descriptive analysis to compare acquisition characteristics and performance indicators, including revenue, production output, operating costs, and earnings per share when available.
4. Evaluate acquisition outcomes while considering broader market factors, such as changes in oil prices and industry conditions.
5. Develop a logistic regression classification model using pre-deal financial variables to predict whether acquisitions are more likely to create or destroy value.
6. Evaluate model performance using cross-validation and apply the trained model to four Houston-area acquisitions that were excluded from the training dataset.
7. Present the results through a technical report and business presentation, including model findings, limitations, and recommendations for future improvement.

Deal	Value	Closed
ExxonMobil acquires Pioneer Natural Resources	\$59.5B	2024
ConocoPhillips acquires Marathon Oil	\$22.5B	Nov 2024
Chevron acquires PDC Energy	\$7.6B	Mid-2023
Diamondback Energy acquires Endeavor Energy Resources	\$26B	2024

5. Data Collection and Preprocessing

The dataset used in this analysis was built from deals.csv, a manually curated list of 95 energy-sector M&A transactions compiled by the team. Each entry recorded the deal's ticker, acquirer, target, announcement date, close date, deal value, and, where the ticker was delisted or ambiguous, a manually resolved SEC Central Index Key (CIK). This curation step kept the dataset scoped to genuine, comparable, arm's-length M&A transactions in the energy sector rather than every filer the ticker map happened to return.

Field	Purpose
ticker, acquirer, target	Deal identification

announce_date, close_date	Anchors for the pre- and post-merger quarter windows
deal_value_usd_b	Deal size, used later as a model feature
cik	SEC Central Index Key; pre-supplied for 68 of 95 deals, the remaining 27 resolved live from SEC's ticker map
notes	Team curation flags, including confidence level and items flagged for team discussion before inclusion

Using this curated list as the backbone, the team built a scraper that made automated API calls to the SEC's free, public companyfacts endpoint (data.sec.gov/api/xbrl/companyfacts/) for each acquirer, pulling every quarterly financial fact filed in that company's 10-Q and 10-K filings. For each deal, the scraper collected 8 quarters of data immediately before the announcement date and 8 quarters immediately after the close date; quarters falling between announcement and close were excluded, since the acquirer is mid-transaction and neither window applies cleanly. This process produced quarterly_long.csv, containing one row per deal per quarter per financial ratio (ROA, ROE, NPM, GPM, D/E, and EPS).

Ratio calculations:

Each quarter's six ratios are computed directly from the underlying XBRL line items (net income, revenue, gross profit, diluted EPS, total assets, stockholders' equity, total liabilities, and long-term debt) using the formulas below.

Ratio	Formula	Notes
ROA	$\frac{(\text{Net Income} \times 4)}{\text{Total Assets}}$	Net income is annualized ($\times 4$) so a one-quarter flow is comparable in scale to the point-in-time asset balance
ROE	$\frac{(\text{Net Income} \times 4)}{\text{Stockholders' Equity}}$	Same annualization logic as ROA
NPM	$\text{Net Income} \div \text{Revenue}$	Both terms are flow measures over the same quarter, so no annualization is needed
GPM	$\text{Gross Profit} \div \text{Revenue}$	Same as NPM; excluded from modeling due to 96% missingness (Section 6)

D/E	Total Liabilities ÷ Stockholders' Equity	Falls back to Long-Term Debt ÷ Equity when a company does not file a Total Liabilities figure
EPS	Diluted Earnings Per Share	Taken directly from the XBRL EarningsPerShareDiluted fact; not derived from other fields

The quarterly data was then collapsed into `features_wide.csv`, one row per deal, by averaging each of the six ratios across its pre-merger window: `pre_ROA_mean`, `pre_ROE_mean`, `pre_NPM_mean`, `pre_GPM_mean`, `pre_DE_mean`, and `pre_EPS_mean` each represent the mean of that ratio's available pre-merger quarterly values (up to the 8-quarter maximum). Two companion fields, `n_pre_quarters` and `n_post_quarters`, record how many usable quarters actually went into that average for each deal, since gaps in a company's SEC filing history mean not every deal has the full 8 quarters on both sides; `complete_post_window` flags whether the full post-merger window was available. Each deal was also assigned a binary outcome label (`draft_label`) indicating whether the merger was classified as value-creating or value-destroying, described further in Section 8. Although `features_wide.csv` computes averaged pre-deal values for all six ratios, the modeling step in Section 6 retained only pre-deal ROA, D/E, and EPS, together with deal value, as predictors.

Of the 95 curated deals, 85 were successfully scraped; the remainder failed at the CIK-resolution or XBRL-fetch step. Missing data was handled in two ways: GPM was excluded entirely from the feature set due to high missingness across the sample, and for the remaining features, median rather than mean imputation was used, since a small number of extreme outliers would otherwise distort the imputed values (see Section 6 for the resulting median values used).

6. Feature Engineering

An outlier check flagged Chesapeake Energy's acquisition of Vine Energy, with a pre-deal EPS of -150.26 , far outside the range of every other deal. To avoid this single observation distorting imputed values, the team used **median** rather than mean imputation:

Feature	Median used
<code>pre_ROA_mean</code>	0.0209
<code>pre_DE_mean</code>	0.9982
<code>pre_EPS_mean</code>	0.2900
<code>deal_value_usd_b</code>	4.9000

The four selected features were standardized using StandardScaler so that each variable had a mean close to 0 and a standard deviation close to 1. This made it easier to compare the logistic regression coefficients across variables.

7. Exploratory Data Analysis (EDA)

Pre-deal characteristics by outcome:

Feature	Destroying — mean	Creating — mean	Destroying — median	Creating — median
Pre-Deal ROA	-0.022	-0.018	0.035	0.018
Pre-Deal D/E Ratio	0.563	3.057	0.957	1.040
Pre-Deal EPS	-5.615	0.435	0.350	0.214
Deal Value (\$B)	8.453	5.403	5.0	4.3

The exploratory analysis showed that some differences between the two groups were influenced by extreme observations rather than representing a consistent pattern across all deals. For example, EPS showed a large difference in mean values, but the median values were much closer, suggesting that a few financially distressed companies significantly affected the average. A similar pattern was observed for the debt-to-equity ratio, where a small number of highly leveraged deals influenced the mean difference between groups.

The correlation analysis showed a moderate relationship between Pre-Deal ROA and Pre-Deal EPS ($r = 0.58$), which is expected because both measures are related to company profitability. However, the correlation was not high enough to remove either variable from the model.

None of the four features showed a strong individual relationship with the outcome label. Debt-to-equity ratio had the strongest correlation ($r = 0.220$), followed by EPS ($r = 0.130$), Deal Value ($r = -0.146$), and ROA ($r = 0.010$). These results suggested early on that the selected financial features provided limited predictive signal for distinguishing value-creating and value-destroying deals.

8. Modeling Techniques

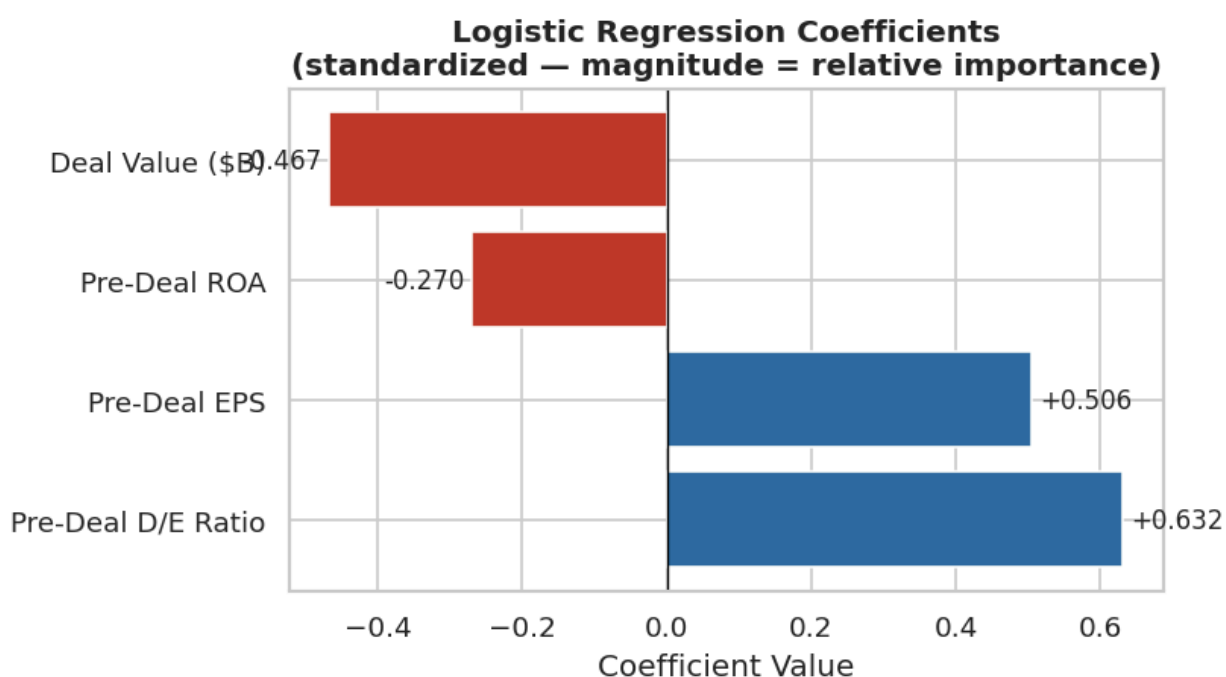
Logistic Regression: Chosen because it is easy to interpret and works well for binary classification problems.

Regularization: The default L2 regularization with **C = 1.0** was used to help reduce the risk of overfitting, especially since the dataset only contained 56 observations.

Model Validation: A **5-fold stratified cross-validation** approach was used so that each fold maintained a similar proportion of value-creating and value-destroying acquisitions.

Final coefficients (fit on the full labeled sample), intercept = -0.0873 :

Feature	Coefficient	Odds ratio	Direction
Pre-Deal D/E Ratio	+0.632	1.88	Higher pre-deal leverage → higher odds of value creation
Pre-Deal EPS	+0.506	1.66	Stronger pre-deal profitability → higher odds of value creation
Pre-Deal ROA	-0.270	0.76	Higher ROA → lower odds of value creation
Deal Value (\$B)	-0.467	0.63	Larger deals → lower odds of value creation



The logistic regression coefficients show how each feature influenced the model's prediction. Pre-deal debt-to-equity ratio had the strongest positive relationship with value creation, with an odds ratio of 1.88, suggesting that higher leverage was associated with higher predicted odds of

value creation. Pre-deal EPS also had a positive effect, with an odds ratio of 1.66, indicating that stronger profitability was linked to better predicted outcomes.

In contrast, pre-deal ROA and deal value had negative coefficients. The model associated higher ROA with lower odds of value creation, which is counter-intuitive and may reflect patterns specific to this limited dataset rather than a true economic relationship. Deal value had an odds ratio of 0.63, suggesting that larger acquisitions were less likely to be classified as value-creating.

These coefficients help explain how the model makes predictions, but they should be interpreted cautiously because the model did not demonstrate strong predictive performance.

9. Model Evaluation Metrics

Cross-validated performance:

Metric	Mean	Std. dev. across folds
Accuracy	0.497	± 0.164
AUC-ROC	0.522	± 0.174
Precision	0.600	± 0.255
Recall	0.373	± 0.110

The cross-validation results show that the model achieved an average accuracy of 49.7% and an AUC-ROC of 0.522, indicating performance close to random guessing. The large variation across folds ($\pm 16.4\%$ accuracy and $\pm 17.4\%$ AUC) suggests that the model is unstable due to the small sample size.

The model achieved higher precision (0.600) but lower recall (0.373), meaning it was more accurate when making positive predictions but missed many actual value-creating deals. Overall, the results indicate that the current model has limited predictive ability and would require additional data and stronger features for improvement.

Classification report (pooled cross-validated predictions, n = 56):

```

=== Classification Report (Cross-Validated) ===
              precision    recall  f1-score   support

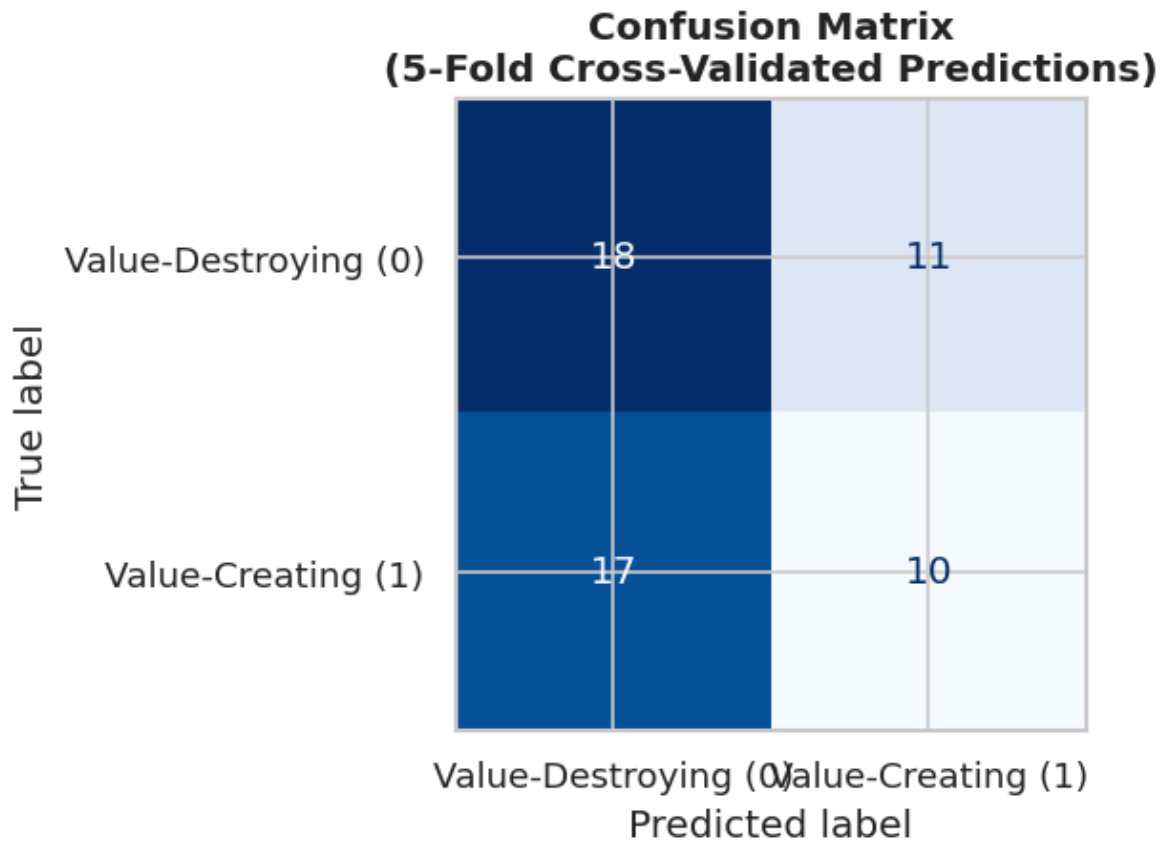
Value-Destroying      0.51      0.62      0.56         29
Value-Creating        0.48      0.37      0.42         27

   accuracy                   0.50         56
  macro avg              0.50      0.50      0.49         56
 weighted avg            0.50      0.50      0.49         56

```

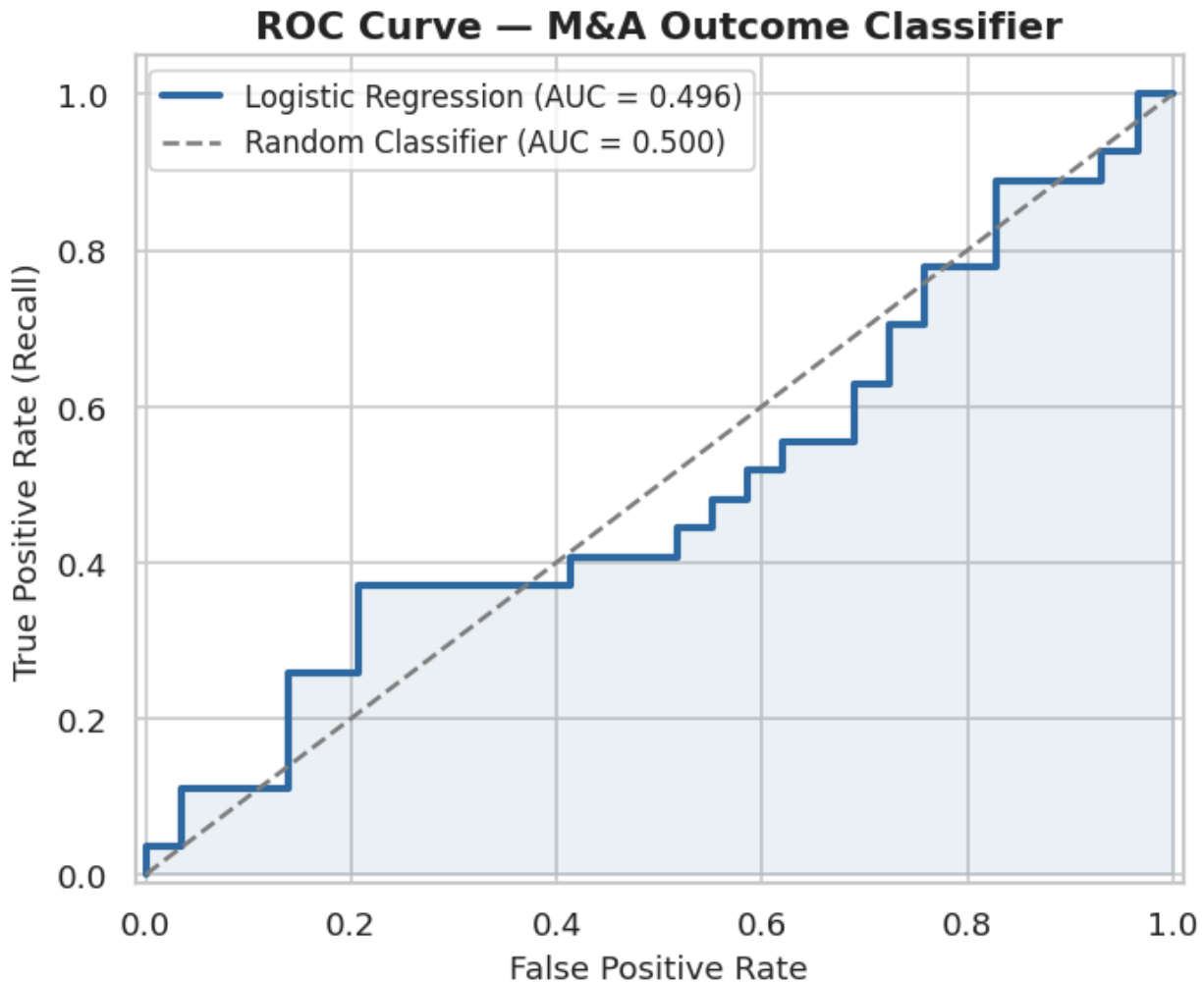
The cross-validated classification report shows that the logistic regression model achieved an overall accuracy of 50%, correctly classifying 28 out of 56 observations. The model performed slightly better in identifying value-destroying deals, with a precision of 0.51, recall of 0.62, and F1-score of 0.56. In comparison, the value-creating class had lower performance, with precision 0.48, recall 0.37, and F1-score 0.42.

The results show that the model performed slightly better when identifying value-destroying deals but struggled to detect value-creating acquisitions. The similar macro and weighted F1-scores (approximately 0.49) indicate that the model did not achieve strong separation between the two classes.



Among the 29 value-destroying deals, the model correctly classified 18 cases and incorrectly classified 11 cases as value-creating. For the 27 value-creating deals, the model correctly identified only 10 cases while misclassifying 17 as value-destroying.

These results show that the model was slightly better at identifying value-destroying deals but struggled to recognize deals that actually created value. The number of incorrect predictions in both categories suggests that the current set of financial features does not provide enough information for the model to separate the two outcome groups. This limitation is likely related to the small sample size and the lack of additional operational and market-related variables.



The ROC curve was used to evaluate how well the model could distinguish between value-creating and value-destroying acquisitions under different classification thresholds. The model achieved an AUC of 0.496, which is almost the same as random guessing (AUC = 0.50). The ROC curve also stays close to the diagonal reference line, showing that the model has limited ability to separate the two outcome groups.

10. Application of the Model to Houston Showcase Deals

```

=== Houston Deal Predictions ===
    acquirer          target  deal_value_usd_b  Predicted Outcome  P(Value-Creating)
    Chevron          PDC Energy           7.6  Value-Destroying X           0.429
    ExxonMobil Pioneer Natural Resources    64.5  Value-Destroying X           0.055
    Diamondback Energy      Endeavor Energy    26.0  Value-Destroying X           0.247
    ConocoPhillips      Marathon Oil        22.5  Value-Destroying X           0.271
  
```

Although the model showed limited predictive performance, it was still applied to the four Houston-area acquisitions to demonstrate how the framework could be used with unseen transactions. These deals were excluded from training to simulate how the model would

evaluate future unseen transactions. The model classified all four Houston-area acquisitions as value-destroying, although the prediction confidence differed across the deals. Among the four, Chevron and PDC Energy were closest to the decision boundary, with a 43% predicted probability of being value-creating. In contrast, ExxonMobil and Pioneer Natural Resources received the lowest probability of value creation at 5.5%. This prediction was largely influenced by the deal's exceptionally large transaction value, which had the strongest negative effect in the logistic regression model.

These predictions should be viewed as an illustration of the model's application rather than reliable conclusions about the success or failure of the acquisitions. If the model is improved with a larger dataset and more informative financial and operational variables, the same prediction process could provide more meaningful decision support for future M&A transactions.

11. Insights from the Results

The model coefficients suggest that higher EPS and debt-to-equity ratios were associated with higher predicted probabilities of value creation within this sample. However, these relationships should be interpreted cautiously because the model performance was close to random guessing and may reflect patterns specific to this sample rather than true economic relationships.

The exploratory data analysis also showed that a few extremely large acquisitions influenced the average values of several financial variables. When median values were compared instead, the differences between value-creating and value-destroying deals became much smaller. This suggests that the dataset contains outliers that can affect the interpretation of the results.

Finally, the model appeared to rely heavily on deal value when evaluating the four Houston area acquisitions, particularly the ExxonMobil and Pioneer Natural Resources transactions. This indicates that the current model may be placing too much emphasis on deal size rather than capturing the overall financial characteristics of the acquisition.

12. Actionable Recommendations

One way to improve this project is to make the data collection process more reproducible by integrating SEC EDGAR data directly into the workflow. This would allow financial information to be collected and updated more consistently over time.

The model could also be strengthened by including additional measures of acquisition performance, such as changes in revenue, production levels, operating costs, and earnings per share after the acquisition. These variables may provide a more complete picture of whether a deal creates value.

It would also be beneficial to consider broader market conditions, particularly oil prices and overall energy industry trends. Including these factors could help distinguish the effects of an acquisition from changes caused by the external market.

Another area for improvement is the creation of the outcome variable. Carefully reviewing and documenting how the target label is defined would improve the reliability of the model, since prediction accuracy depends on the quality of the target variable. Future research could also compare logistic regression with other machine learning methods, such as regularized regression or tree-based models, to determine whether they capture additional patterns in the data.

Finally, expanding the dataset to include more historical energy M&A transactions would improve the stability of the model and provide more reliable estimates during cross-validation.

13. Limitations of the Study

This study has several limitations that should be considered when interpreting the results.

- The model only used four pre-deal financial variables and did not include important post-acquisition performance indicators such as revenue growth, production changes, or operating cost improvements. As a result, it cannot fully evaluate acquisition success.
- The analysis also did not account for external factors such as changes in oil prices or overall market conditions, both of which may have influenced acquisition performance.
- In addition, the model achieved an AUC close to 0.50, indicating that its predictive ability was only slightly better than random guessing. Therefore, the predictions should not be used as reliable indicators of acquisition outcomes.
- Finally, the four selected financial variables were chosen because they were available and contained relatively few missing values. Other financial or operational variables may provide stronger predictive power.

14. Future Work and Enhancements

Future research should focus on improving both the data and the predictive model. First, the complete SEC EDGAR data collection process should be integrated into the project to create a fully reproducible workflow.

Second, additional financial and operational variables should be included, along with pre- and post-acquisition performance analysis, to better measure the impact of each transaction.

Third, the methodology used to define value-creating and value-destroying acquisitions should be validated and clearly documented.

Finally, future studies should expand the dataset by including more acquisitions and evaluate more advanced machine learning models. With a larger, more comprehensive dataset, the prediction framework demonstrated in this project could become a more useful decision-support tool for evaluating future energy-sector mergers and acquisitions.

15. Conclusion

This project developed a proof-of-concept machine learning framework for evaluating whether energy-sector M&A transactions may create or destroy value using information available before a deal closes. Although the logistic regression model did not achieve strong predictive performance, the project successfully demonstrated the complete analytical process, including data preparation, feature engineering, model development, cross-validation, and testing on unseen Houston-area acquisitions.

The results suggest that four pre-deal financial variables alone are not sufficient to reliably predict M&A outcomes. The near-chance AUC performance indicates that acquisition success depends on more complex factors than the financial characteristics included in the current model. However, this result provides a useful baseline for future development rather than indicating that the approach is ineffective.

Future improvements should focus on expanding the dataset, adding operational and market-related variables, improving the outcome-label methodology, and testing alternative modeling approaches. With additional data and stronger features, this framework could become a more useful decision-support tool for evaluating future energy-sector acquisitions.